

NLP OUTPUTS:

EXPERIMENT 1:

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PS C:\Users\dhava\OneDrive\Desktop\NLP> & 'C:\Users\dhava\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\dhava\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle\libs\debugpy\launcher' '51543' '--' 'C:\Users\dhava\OneDrive\Desktop\NLP\experiment1.py'
[nltk_data] Downloading package punkt to
[nltk_data]   C:\Users\dhava\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to
[nltk_data]   C:\Users\dhava\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
[nltk_data] Downloading package wordnet to
[nltk_data]   C:\Users\dhava\AppData\Roaming\nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
Enter a sentence:I am thinking that I will becoming the most.

Original Text:
I am thinking that I will becoming the most.

Tokens:
['I', 'am', 'thinking', 'that', 'I', 'will', 'becoming', 'the', 'most', '.']

Stemmed Words:
['i', 'am', 'think', 'that', 'i', 'will', 'becom', 'the', 'most', '.']

Lemmatized Words:
['I', 'am', 'thinking', 'that', 'I', 'will', 'becoming', 'the', 'most', '.']

Comparison:
Stemming reduces words to root forms, which may not be meaningful.
Lemmatization converts words to meaningful base forms.
PS C:\Users\dhava\OneDrive\Desktop\NLP> █
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EXPERIMENT 2:

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PS C:\Users\dhava\OneDrive\Desktop\NLP> c:; cd 'C:\Users\dhava\OneDrive\Desktop\NLP'; & 'C:\Users\dhava\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\dhava\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle\libs\debugpy\launcher' '51455' '--' 'C:\Users\dhava\OneDrive\Desktop\NLP\experiment2.py'
[nltk_data] Downloading package punkt to
[nltk_data]   C:\Users\dhava\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to
[nltk_data]   C:\Users\dhava\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
[nltk_data] Error loading averaged_perceptron_tagged_eng: Package
[nltk_data]   'averaged_perceptron_tagged_eng' not found in index
Enter a sentence:I am the most dangerous.

Tokens:
['I', 'am', 'the', 'most', 'dangerous', '.']

POS Tags:
I -> PRP
am -> VBP
the -> DT
most -> RBS
dangerous -> JJ
. -> .

Tag Meanings:
NN -> Noun
VB -> Verb
JJ -> Adjective
RB -> Adverb
PRP -> Pronoun
DT -> Determiner

Total Words: 6
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EXPERIMENT 3:

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PS C:\Users\dhava\OneDrive\Desktop\NLP> c++; cd "c:\Users\dhava\OneDrive\Desktop\NLP"; & "C:\Users\dhava\AppData\Local\Programs\Python\Python313\python.exe" "c:\Users\dhava\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher" "49727" "--" "C:\Users\dhava\OneDrive\Desktop\NLP\exp3.py"
[nltk_data] Downloading package wordnet to
[nltk_data]   C:\Users\dhava\AppData\Roaming\nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
[nltk_data] Downloading package omw-1.4 to
[nltk_data]   C:\Users\dhava\AppData\Roaming\nltk_data...
[nltk_data]   Package omw-1.4 is already up-to-date!
Enter number of headlines: 4
Enter headline 1: Stock market gains record profits
Enter headline 2: Investors celebrate rise in share prices
Enter headline 3: Football team wins championship
Enter headline 4: Cricket tournament attracts audience

Cosine Similarity Matrix:
[[1. 0. 0. 0.]
 [0. 1. 0. 0.]
 [0. 0. 1. 0.]
 [0. 0. 0. 1.]]

Headline Clusters:
Stock market gains record profits -> Cluster 0
Investors celebrate rise in share prices -> Cluster 0
Football team wins championship -> Cluster 0
Cricket tournament attracts audience -> Cluster 1

Enter first word: Cricket
Enter second word: wins
WordNet Similarity: 0.05263157894736842
PS C:\Users\dhava\OneDrive\Desktop\NLP>
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EXPERIMENT 4:

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PS C:\Users\dhava\OneDrive\Desktop\NLP> c++; cd "c:\Users\dhava\OneDrive\Desktop\NLP"; & "C:\Users\dhava\AppData\Local\Programs\Python\Python313\python.exe" "c:\Users\dhava\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher" "49803" "--" "C:\Users\dhava\OneDrive\Desktop\NLP\exp4.py"
Enter number of documents: 3
Enter document: Machine learning improves healthcare diagnosis
Enter document: Deep learning is used in artificial intelligence
Enter document: Climate change affects intelligence

Enter search query: artificial intelligence

TF-IDF Similarity Scores:
Document 1 : 0.0
Document 2 : 0.506
Document 3 : 0.243

LSA Similarity Scores:
Document 1 : 0.248
Document 2 : 0.942
Document 3 : 0.87

Most Relevant Document:
Deep learning is used in artificial intelligence
PS C:\Users\dhava\OneDrive\Desktop\NLP>
```

EXPERIMENT 5:

```
PS C:\Users\dhava\OneDrive\Desktop\NLP> c++; cd "c:\Users\dhava\OneDrive\Desktop\NLP"; & "C:\Users\dhava\AppData\Local\Programs\Python\Python313\python.exe" "c:\Users\dhava\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher" "58505" "--" "C:\Users\dhava\OneDrive\Desktop\NLP\exp5.py"
[nltk_data] Downloading package punkt to
[nltk_data]   C:\Users\dhava\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to
[nltk_data]   C:\Users\dhava\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data]   C:\Users\dhava\AppData\Roaming\nltk_data...
[nltk_data]   Package averaged_perceptron_tagger_eng is already up-to-date!
Enter legal text: John Smith filed a case against ABC corporation in New York court

Detected Named Entities:
John ->ENTITY
Smith ->ENTITY
ABC ->ENTITY
New ->ENTITY
York ->ENTITY

Enter actual number of entities: 7
NER Accuracy: 71.43 %
PS C:\Users\dhava\OneDrive\Desktop\NLP>
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EXPERIMENT 6:

```
PS C:\Users\dhava\OneDrive\Desktop\NLP> c;; cd 'c:\Users\dhava\OneDrive\Desktop\NLP'; & 'C:\Users\dhava\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\dhava\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle\libs\debugpy\launcher' '54419' '--' 'C:\Users\dhava\OneDrive\Desktop\NLP\exp6.py'
[nltk_data] Downloading package punkt to
[nltk_data]   C:\Users\dhava\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to
[nltk_data]   C:\Users\dhava\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
Enter biomedical sentence: Aspirin reduces fever in patients
Actual Relation(1/0):1

Tokens:
['aspirin', 'reduces', 'fever', 'in', 'patients']

Predicted Relation: 1
Precision: 1.0
Recall: 1.0
F1-Score: 1.0
PS C:\Users\dhava\OneDrive\Desktop\NLP> █
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EXPERIMENT 7:

```
PS C:\Users\dhava\OneDrive\Desktop\NLP> & 'C:\Users\dhava\AppData\Local\Programs\Python\Python313\python.exe' 'c:\Users\dhava\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle\libs\debugpy\launcher' '57622' '--' 'C:\Users\dhava\OneDrive\Desktop\NLP\exp7.py'
[nltk_data] Downloading package punkt to
[nltk_data]   C:\Users\dhava\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to
[nltk_data]   C:\Users\dhava\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
Enter a tweet: AI is transforming social media

Tokens:
['ai', 'is', 'transforming', 'social', 'media']

Unigrams:
[('ai',), ('is',), ('transforming',), ('social',), ('media',)]

Bigrams:
[('ai', 'is'), ('is', 'transforming'), ('transforming', 'social'), ('social', 'media')]

Trigrams:
[('ai', 'is', 'transforming'), ('is', 'transforming', 'social'), ('transforming', 'social', 'media')]

Word Frequencies:
ai : 1
is : 1
transforming : 1
social : 1
media : 1

HMM Prediction (Sample)
AI -> NOUN
improves -> VERB
technology -> NOUN
PS C:\Users\dhava\OneDrive\Desktop\NLP> █
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